



COMSATS University Islamabad (CUI)

Project Proposal

for

Seevia

Version 1.0

By

Malaika Junaid

CIIT/FA22-BAI-020/ISB

Syeda Aleeza Tahir

CIIT/ FA22-BAI-038/ISB

Supervisor

Dr. Samera Batool

Bachelor of Science in Artificial Intelligence (2022-2026)

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Project Category:

☐ A -Desktop Application/Information System	☐ B -Web Application/Web Application based Information System	
☒ C -Problem Solving and Artificial Intelligence	☐ D -Simulation and Modeling	☒ E -Smartphone Application
☐ F -Smartphone Game	☐ G -Networks	☒ H -Image Processing
☐ Other (specify category) _____		

Abstract

Seevia is an innovative AI-powered smart assistant designed to empower visually impaired individuals by supporting their independence, safety, and daily living through voice-driven, accessible technology. Visually impaired people often face significant challenges in performing routine tasks such as shopping, navigating stores, managing household inventory, and handling emergencies due to the limited availability of integrated, user-friendly assistive tools. Current assistive technologies are either fragmented or not tailored to local languages and cultural needs, especially in regions like Pakistan. This project seeks to address the lack of a unified, real-time solution that allows blind or low-vision users to interact with their surroundings without constant human assistance. It also targets limitations in affordability, personalization, and language inclusivity found in existing systems. There is no context-aware, AI-enabled assistant available that combines visual recognition, voice interaction, and task automation for visually impaired individuals in developing countries. Seevia will offer features such as product identification, currency recognition, pantry management, and emergency support through voice and camera-based interaction. It aims to provide real-time feedback and personalized experiences using user behavior learning. The assistant will support multilingual commands and intuitive interaction to ensure usability across diverse user groups. This solution will bridge accessibility gaps by delivering a localized, scalable, AI-intensive product that enables independence, reduces reliance on caregivers, and improves the quality of life for visually impaired users.

1. Introduction

The purpose of this project proposal document is to outline the development and objectives of Seevia, an AI-powered smart assistant designed specifically for visually impaired individuals. Seevia aims to address the challenges faced by visually impaired individuals in completing everyday tasks such as shopping, navigating stores, and tracking household inventory. The app will integrate AI, voice recognition, and camera-based input to offer a hands-free, user-friendly experience. It will feature product recognition, pantry tracking, emergency alerts, and personalized user profiles.

Visually impaired individuals often lack accessible tools to help them with routine activities, leading to dependence on others. Existing assistive technologies are fragmented, lacking voice control, integration, and personalization. Seevia will fill this gap by offering an all-in-one, AI-enabled assistant that empowers users to independently manage their daily tasks. The app will support voice commands and real-time product recognition, making it easier for users to perform essential tasks without external help.

Seevia will learn from user behavior and adapt over time, improving personalization and responsiveness. This adaptability will ensure greater safety, self-sufficiency, and confidence for visually impaired users. The app will promote autonomy by reducing reliance on caregivers and helping users make informed decisions independently. Additionally, the emergency support feature will provide immediate assistance when required.

This proposal outlines the technical, operational, and developmental aspects of Seevia, aiming to enhance the quality of life for visually impaired individuals by providing a seamless, accessible platform that integrates essential functions into one comprehensive solution.

2. Problem Statement

Visual impairment significantly impacts the daily lives of millions around the world, creating barriers to independence and everyday functionality. According to the World Health Organization (WHO), over 2.2 billion people globally suffer from some form of visual impairment, with at least 1 billion cases being preventable or untreated. In Pakistan, about 2.7% of adults over 30 are blind, while 1.7% live with low vision, affecting over 1.8 million people. Many individuals are born with visual impairments, while others acquire them due to conditions such as cataracts, glaucoma, diabetes, or injury. Although treatments like corrective lenses or surgeries exist, a significant number of individuals remain untreated due to lack of access, financial hardship, or unawareness.

As a result, many visually impaired individuals are forced to depend on caregivers for routine tasks. They often struggle with basic activities like navigating unfamiliar environments, recognizing products, managing household items, or handling shopping independently. These limitations restrict their participation in society and reduce their quality of life. While assistive technologies do exist, they are often expensive, inaccessible, or fragmented in functionality. Most lack voice integration, contextual understanding, or the ability to provide dynamic, real-time assistance. The absence of a unified, intelligent, and affordable solution leaves users without effective tools to manage daily challenges.

This dependency leads to a loss of confidence and autonomy, affecting emotional well-being and personal development. Children and young people with vision impairments experience these challenges early, often facing isolation or exclusion. The lack of independence also affects educational and employment opportunities. Many visually impaired individuals report feelings of helplessness due to their inability to manage simple tasks on their own. The gap between available tools and real-life needs remains wide, especially in developing countries. There is an urgent need for a more inclusive and adaptable technological solution that can help visually impaired individuals lead self-directed, independent lives.

3. Problem Solution/Objectives of the Proposed System

To address the pressing challenges faced by visually impaired individuals in daily life, **Seevia** is proposed as a comprehensive AI-powered assistant. This mobile application is designed to facilitate independent living by enabling users to perform routine tasks through voice commands and camera input. The system will provide a unified platform for managing household chores, product identification, emergency alerts, and personalized assistance. It eliminates the need for multiple standalone tools by offering an all-in-one solution tailored for accessibility. Through natural language processing (NLP), Seevia will interpret user commands and offer real-time responses.

The application's product recognition feature will allow users to identify items while shopping or at home by simply pointing their phone's camera. Its smart pantry module will help track groceries and notify users when stock is low or expiring. In emergency situations, the app will offer a one-tap or voice-activated SOS feature to contact caregivers or emergency services. Personalized profiles will allow the app to adapt to each user's preferences, routines, and accessibility needs. Over time, Seevia will use machine learning to optimize its recommendations and interactions based on user behavior.

The application will be developed for Android devices to ensure maximum accessibility and affordability. Its interface will be simple, intuitive, and optimized for screen reader compatibility. Users with minimal technological proficiency will be able to navigate the app with ease. Seevia will also contribute to psychological well-being by helping users regain confidence and reduce their dependence on others. Emotional empowerment is as important as functional support in the app's design philosophy. The system aims to support users in living a life of dignity, independence, and improved quality. The solution goes beyond basic assistance by offering a dynamic platform that learns and evolves. Ultimately, Seevia aims to bridge the accessibility gap using artificial intelligence, improving the lives of those often overlooked by mainstream technology.

3.1 Objectives

BO-1: Enable visually impaired individuals to perform daily tasks independently using an AI-powered voice assistant.

BO-2: Enhance user autonomy by integrating personalized features for pantry management, shopping assistance, and emergency support.

BO-3: Improve user safety by providing real-time emergency alerts and easy access to emergency contacts or help.

BO-4: Promote self-esteem and confidence in visually impaired users by reducing their dependence on caregivers for routine activities.

BO-5: Improve the overall quality of life for visually impaired individuals by providing a comprehensive, accessible mobile platform that simplifies day-to-day tasks.

4. Related System Analysis/Literature Review

Existing systems for visually impaired individuals offer partial solutions but often have limitations in terms of functionality, accessibility, or cost. While some focus on navigation or product recognition, they lack integration of essential daily tasks like shopping assistance, pantry management, and emergency support. Others rely on external help, leading to variability in service. Seevia aims to provide a comprehensive, autonomous AI-powered solution that addresses these gaps and enhances user independence.

Table 1 Related System Analysis with proposed project solution

Application Name	Weakness	Proposed Project Solution
Aira	Relies on human agents, subscription-based service, can be costly and less accessible.	Seevia is fully AI-powered, removing the need for human agents, offering cost-effective, autonomous solutions.
Be My Eyes	Depends on volunteers, leading to variability in service availability.	Seevia eliminates the need for external volunteers, offering a reliable and continuous AI-powered assistant.
Seeing AI	Lacks integration for household management and emergency support.	Seevia combines AI, voice control, and camera recognition to offer shopping assistance, pantry management, and emergency alerts in one platform.
TellBack	Focuses only on text-to-speech and basic navigation, lacking task-specific tools.	Seevia enhances TalkBack's functionality by adding product recognition, personalized task management, and emergency support through an integrated AI assistant.
OrCam MyEye	Expensive hardware solution, not accessible to all visually impaired users.	Seevia provides a more affordable software solution through smartphones, making it accessible to a larger group of users.

5. Vision Statement

For visually impaired individuals **who** face challenges in performing daily tasks independently due to limited access to consistent and intelligent assistance, **the** Seevia **is** an AI-powered smart assistant application **that** provides voice-controlled task execution, object and product recognition, shopping and pantry management, and emergency support. **Unlike** solutions that rely on human agents, volunteers, or expensive hardware, **our product** (Seevia) offers an affordable, fully autonomous, and integrated AI solution through smartphones, empowering users to navigate daily life with confidence and independence.

6. Scope

The scope of Seevia is to design and develop an AI-powered mobile application tailored specifically for visually impaired individuals to assist them in performing daily activities independently and efficiently. The application will include multiple integrated modules that address essential aspects of everyday life through voice-controlled interaction, camera-based recognition, and intelligent decision-making systems. Core functionalities will include a voice command interface for hands-free operation, enabling users to control the app through natural speech. A personalized user profile will store preferences, health data, and app settings for a customized experience. The pantry and shopping management module will allow users to track household items and generate shopping lists through barcode scanning and voice inputs. A shopping assistant will provide real-time product information using object detection and OCR technologies. The system will also feature an emergency alert mechanism that allows users to trigger help requests via voice or gesture input.

Additional functionalities may include navigation assistance within limited spaces, calendar/task reminders, and smart suggestions based on behavioral data. The app will be optimized for offline functionality for critical tasks and designed to be compatible with low- to mid-range smartphones for broader accessibility. The project will utilize deep learning models, NLP, and computer vision techniques to recognize objects, interpret commands, and make context-aware decisions. Data privacy and user safety will be ensured through secure authentication and encryption techniques. The system boundaries are defined to exclude general-purpose navigation (e.g., Google Maps-style outdoor routing) and complex medical diagnostics. Instead, the focus remains on assisting with domestic and personal tasks. The solution will be AI-driven and self-sufficient, minimizing reliance on human agents or external devices. Integration with third-party services may be included for voice APIs, barcode databases, or cloud storage, but the app will function independently of external assistance. The final product will be a fully functional prototype, tested with feedback from real users to ensure its practicality and effectiveness. In essence, Seevia aims to become a dependable digital companion that enables visually impaired individuals to live more confidently and independently.

7. Modules

7.1 Module 1: User Profile & Settings

FE-1: Create and manage user profiles with voice-guided onboarding and personalization.

FE-2: Customize app settings including voice pitch, command sensitivity, language, and accessibility preferences (e.g., high-contrast mode, voice-only interaction).

FE-3: Store and manage basic health information such as medications, allergies, and emergency notes.

7.2 Module 2: Voice Command Interface

FE-1: Activate and control app features using voice commands.

FE-2: Provide voice feedback and prompts for user queries.

FE-3: Interpret natural language inputs for better user interaction.

FE-4: Operate in offline mode for essential commands.

7.3 Module 3: Pantry Management

FE-1: Add items to pantry using barcode scanning or voice input.

FE-2: Track item expiry dates and notify users.

FE-3: Generate shopping lists based on pantry data.

FE-4: Allow voice-based edits and list management.

FE-5: Recommend or restrict items based on user-defined dietary preferences or allergies from profile.

7.4 Module 4: Shopping Assistant & Product Recognition

FE-1: Identify products using camera-based object detection and OCR.

FE-2: Provide product details (name, category, brand, expiry).

FE-3: Assist in finding similar or alternative items.

FE-4: Guide users through shopping process using real-time voice feedback.

FE-5: Learn and store layout maps of shopping centers for improved navigation.

7.5 Module 5: Emergency Management & Unusual Activity Detection

FE-1: Detect and respond to unusual activities (e.g., long periods of inactivity, falls, or abnormal movement patterns) using sensor input and contextual analysis.

FE-2: Send emergency alerts to predefined contacts with GPS location via voice command, gesture, or automatic detection.

FE-3: Trigger auditory or vibration-based alarms manually or through predefined voice gestures for instant assistance.

FE-4: Provide real-time voice instructions for first responders based on stored health info.

FE-5: Auto-escalate to volunteer community or services if primary contact is unreachable.

7.6 Module 6: Volunteer Mode

FE-1: Enable non-visually impaired individuals to register as volunteers for emergency assistance.

FE-2: Notify the nearest available volunteer if the user's emergency contacts are unresponsive.

FE-3: Display current proximity-based cases (if any), allowing volunteers to opt-in and respond based on availability.

FE-4: Allow volunteers to enhance Seevia's learning by mapping environments and verifying product data.

FE-5: Include a rating/reputation system for volunteer responses to maintain trust and quality of assistance.

8. System Limitations/Constraints

LI-1: The system relies on smartphone hardware (camera, microphone, internet), which may limit performance or accessibility for users with older or low-end devices.

LI-2: Object detection and product recognition accuracy may be affected by poor lighting or low-resolution images.

LI-3: The AI assistant may have limited performance in offline mode, restricting access to certain features.

LI-4: Initial training datasets may not cover all regional products or household items, reducing recognition accuracy in certain areas.

LI-5: Emergency alert functionality depends on mobile network or internet connectivity to notify contacts.

LI-6: The application currently supports only the Urdu and English languages, limiting usability for speakers of other regional languages.

9. Data Gathering Approach

For the proposed project, data and requirement gathering will be conducted using a combination of interviews and questionnaires targeted at visually impaired individuals, their caregivers, and accessibility experts. Observational studies will also be conducted to understand real-life challenges faced during daily tasks. Additionally, feedback from organizations working with the visually impaired will be collected to ensure the solution is relevant and user-centered. This multi-source approach will help validate user needs and refine system features.

10. Tools and Technologies

Seevia will be built using modern development tools and intelligent systems. This includes technologies for model development, voice processing, computer vision, reinforcement learning, and cross-platform mobile development.

Table 2: Tools and Technologies for Proposed Project

Tools And Technologies	Tools	Version	Rationale
	Android Studio	Giraffe	IDE
	Visual Studio Code	1.88+	IDE
	Firebase	Latest	DBMS
	Figma	CSC 6	Design Work
	GitHub	Git 2.33	Version control
	Technology	Version	Rationale
	Python	3.10	Model development
	TensorFlow Lite	2.14	Model Framework
	OpenCV	4.7	CV Library
	spaCy	3.1	NLP Library
	Hugging Face Transformers	4.36+	NLP Models
	Stable-Baselines3	1.8+	RL Library
	Google ML Kit	Latest	OCR
	Flutter TTS / STT Plugins	Latest	Voice to Text
	Flutter Provider / Riverpod	Latest	State management
	TensorFlow Hub	Latest	Models for Transfer Learning

11. Project Stakeholders and Roles

Table 3 Project Stakeholders for Proposed Project

Project Stakeholder	Role	Responsibility
COMSATS University Islamabad	Project Sponsor	Provides academic oversight, resources, and official support for project development and evaluation.
Malaika Junaid	Co-developer of the system	Responsible for system architecture, core module development, API integration, testing, deployment, and final presentation preparation.
Syeda Aleeza	Co-developer of the system	Responsible for UI/UX design, module integration, stakeholder communication, user testing, research, and documentation.
Dr. Samera Batool	Project Supervisor	Guides the technical direction, reviews milestones, and ensures the project meets academic standards.
Final Year Project Committee	Project Evaluator	Evaluates progress, reviews documentation, and approves the final deliverable.

12. Module based Work Division

Table 4 Team Member Work Division for Proposed Project

Student Name	Student Registration Number	Responsibility/ Module / Feature
Malaika Junaid	CIIT/FA22-BAI-020/ISB	Module 1 – FE 1, 2, 3 Module 2 – FE 1, 2, 3, 4 Module 4 – FE 1, 2, 3, 4, 5
Syeda Aleeza Tahir	CIIT/ FA22-BAI-038/ISB	Module 3 – FE 1, 2, 3, 4, 5 Module 5 – FE 1, 2, 3, 4, 5 Module 6 – FE 1, 2, 3, 4, 5

13. WBS and Gantt Chart

The detailed Work Breakdown Structure (WBS) and Gantt chart outlining tasks, durations, dependencies, and resources are provided in Appendix A.

14. Mockups

The mockups for the major modules of Seevia, as outlined in the Modules section, have been designed to reflect the key functionalities of the application. These include screens for core components such as pantry management, shopping assistant, emergency support, and more. All mockups follow accessibility standards and a voice-first design approach to cater specifically to the needs of visually impaired users.

Mockups are provided in Appendix A.

15. References

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16. Plagiarism Report

 **Similarity Report ID:** oid:30225:463584918

7% Overall Similarity

Top sources found in the following databases:

- 6% Internet database
- 2% Publications database
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TOP SOURCES

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Higher Education Commission Pakistan on 2023-03-17	2%
	Submitted works	
2	Higher Education Commission Pakistan on 2018-10-02	1%
	Submitted works	
3	Stefano Barone, Eva Lo Franco. "Statistical and Managerial Techniques..."	1%
	Crossref	
4	Lindenwood University on 2023-04-16	<1%
	Submitted works	
5	Higher Education Commission Pakistan on 2024-05-22	<1%
	Submitted works	
6	Higher Education Commission Pakistan on 2025-04-29	<1%
	Submitted works	
7	coursehero.com	<1%
	Internet	
8	Higher Education Commission Pakistan on 2019-04-05	<1%
	Submitted works	

Sources overview

Appendix A

Mockups

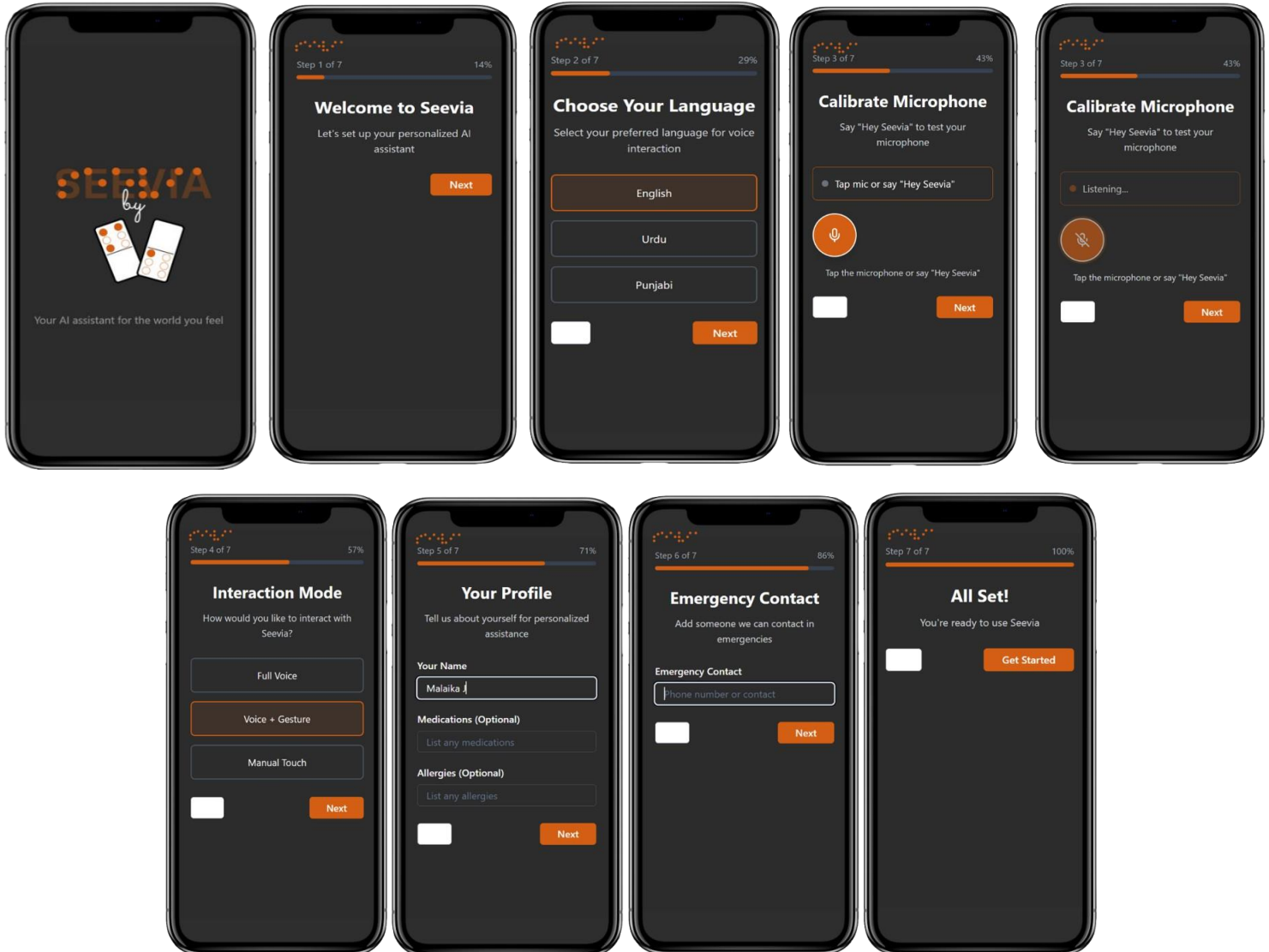


Figure 1 - Onboarding and Profile Screens

Description: Initial setup includes voice-guided onboarding where the user is asked simple questions. Preferences like language, voice pitch, and emergency contacts are saved. This setup is done only once. All fields are accessible via mic input or "Hey Seevia" voice prompts.

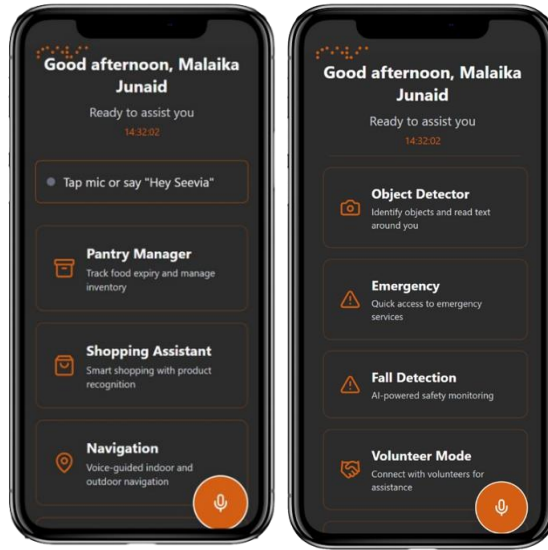


Figure 2 - Dashboard Screen

Description: The main dashboard allows hands-free navigation to all modules. Users can say “Hey Seevia, open pantry” or tap any area to have it read aloud and select it via voice confirmation.

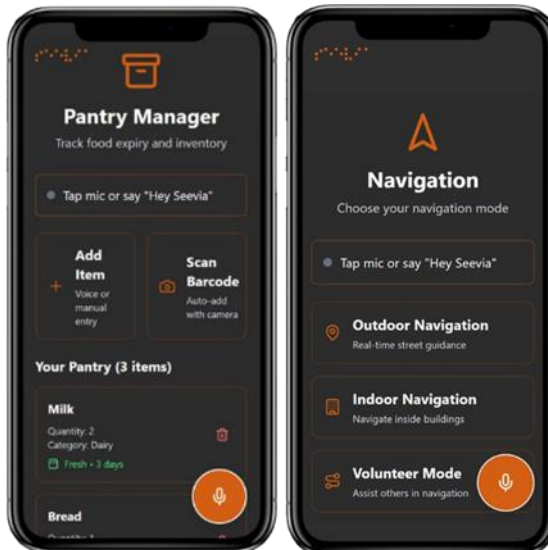


Figure 3 - Pantry Management and Navigation Screen

Description: Users can add items to the pantry by saying “Hey Seevia, add milk,” or scanning barcodes via voice. Navigation through pantry categories is also fully voice-enabled, with automatic audio feedback for current selections.

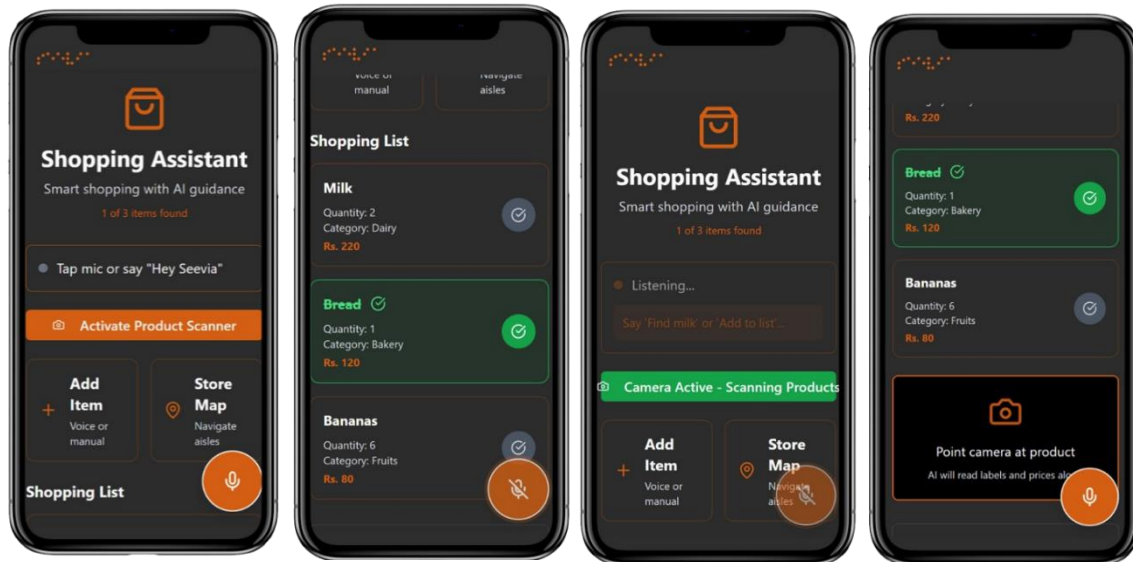


Figure 4 - Shopping Assistant Screen

Description: Voice commands like “Hey Seevia, start shopping” or “add to list” allow seamless interaction. The product scanner identifies items using the camera, marking them as found. Store maps and item categories aid in navigation with voice and visual guidance.

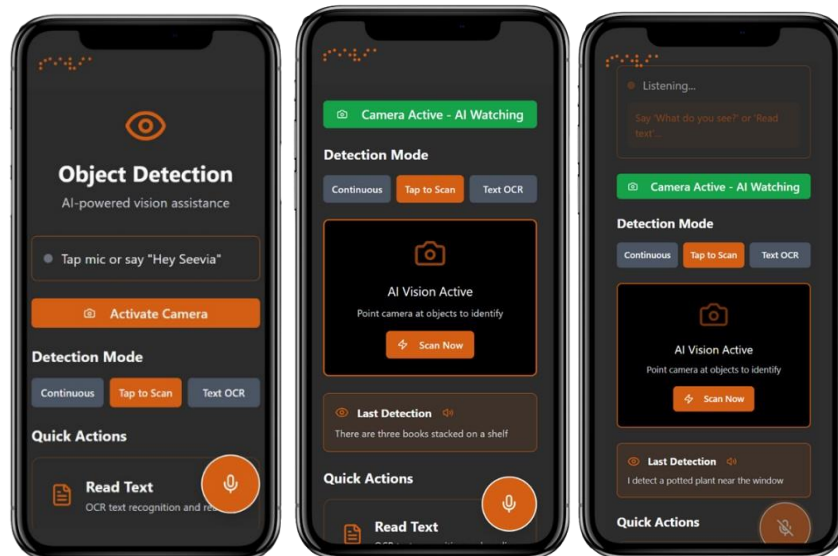


Figure 5 - Object Detection Screen

Description: As the user holds up the phone, “Hey Seevia, what is this?” activates the object detection feature. Recognized product details are read aloud instantly.



Figure 6 - Emergency Mode Screen

Description: Users can say “Hey Seevia, help me”, shake the device, or remain inactive to trigger emergency mode. The app detects falls or unusual inactivity and alerts emergency contacts or volunteers. All actions are voice-controlled with real-time audio feedback.

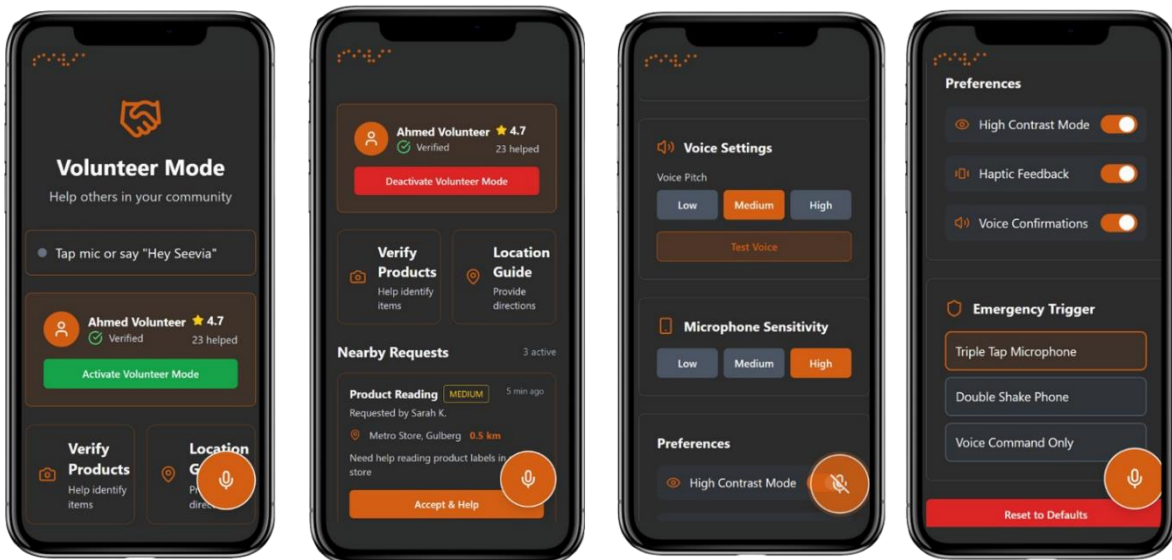


Figure 7 - Volunteer Mode Screen

Description: Designed for non-visually impaired users, these screens support touch-based interaction. Volunteers can view nearby cases, accept help requests, and contribute to environment mapping with quick, intuitive controls. Voice input is optional.

WBS and Gantt chart:

Table 5 Tabular Structure View of WBS for Seevia

Level 1	Level 2	Level 3
Task 1: Analysis	Subtask 1.1: Requirement Meetings	Work Package 1.1.1 – Stakeholder Meetings
		Work Package 1.1.2 – Functional Requirements
	Subtask 1.2: Communication & Research	Work Package 1.2.1 – Stakeholder Communication
		Work Package 1.2.2 – Literature Review
		Work Package 1.2.3 – Feature Finalization
	Subtask 1.3: Wrap-up	Work Package 1.3.1 – Analysis Summary
Task 2: Design	Subtask 2.1: UI/UX Design	Work Package 2.1.1 – Wireframes Per Module
		Work Package 2.1.2 – Accessible UX Guidelines
	Subtask 2.2: System Design	Work Package 2.2.1 – Architecture Flow
		Work Package 2.2.2 – API Interaction Mapping
Task 3: Development	Subtask 3.1: Core Modules	Work Package 3.1.1 – Voice & NLP Module
		Work Package 3.1.2 – Object/Product Detection
		Work Package 3.1.3 – Pantry Management
		Work Package 3.1.4 – Shopping Module
	Subtask 3.2: Context-Aware Modules	Work Package 3.2.1 – Smart Navigation & Mapping
		Work Package 3.2.2 – Emergency + Volunteer
	Subtask 3.3: Base System	Work Package 3.3.1 – Profile & Settings
		Work Package 3.3.2 – System Integration
Task 4: Testing	Subtask 4.1: Internal Testing	Work Package 4.1.1 – Unit Testing
	Subtask 4.2: External Testing	Work Package 4.2.1 – User Testing
		Work Package 4.2.2 – Bug Fixing & Optimization
Task 5: Deployment & Docs	Subtask 5.1: Documentation	Work Package 5.1.1 – Final Report Writing
		Work Package 5.1.2 – Presentation Preparation
	Subtask 5.2: Deployment	Work Package 5.2.1 – App Deployment
Task 6: Completion	Subtask 6.1: Finalization & Maintenance	Work Package 6.1.1 – Project Polishing
		Work Package 6.1.2 – Defense & Evaluation Prep

Table 6 WBS for Seevia

ID	Task	Duration	Resources
1	Analysis	9 d	Malaika Junaid; Syeda Aleeza
2	Requirement Meetings	4 d	Malaika Junaid; Syeda Aleeza
3	Communication with Stakeholders	2 d	Syeda Aleeza
4	Market Research + Literature Review	2 d	Malaika Junaid
5	Feature Finalization	2d	Malaika Junaid; Syeda Aleeza
6	Analysis Finished	1 d	
7	Design	16 d	Malaika Junaid; Syeda Aleeza
8	Sketch Wireframes for Each Module	2 d	Malaika Junaid; Syeda Aleeza
9	UI/UX Design (Voice + Accessibility First)	6 d	Syeda Aleeza
10	System Flow, Architecture Planning	6 d	Malaika Junaid
11	API Flow & Interaction Layer Mapping	7 d	Malaika Junaid; Syeda Aleeza
12	Design Finished	1 d	
13	Development	43 d	Malaika Junaid; Syeda Aleeza
14	Voice & NLP Module	8 d	Syeda Aleeza
15	Object & Product Recognition	7 d	Malaika Junaid
16	Pantry Management	5 d	Syeda Aleeza
17	Shopping Module	3 d	Malaika Junaid
18	Smart Navigation & Environment Learning	9 d	Syeda Aleeza
19	Emergency & Volunteer Modules	16 d	Malaika Junaid
20	Profile & Settings Module	6 d	Malaika Junaid; Syeda Aleeza
21	Development Finished	1 d	
22	Testing	11 d	Malaika Junaid; Syeda Aleeza
23	Unit Testing	5 d	Malaika Junaid
24	User Testing	6 d	Syeda Aleeza
25	Bug Fixes & Refinement	4 d	Malaika Junaid; Syeda Aleeza
26	Testing Finished	1 d	
27	Deployment & Documentation	9 d	Malaika Junaid; Syeda Aleeza
28	Final Report Writing	4 d	Syeda Aleeza
29	App Deployment & Demo	5 d	Malaika Junaid
30	Final Presentation Preparation	4 d	Malaika Junaid; Syeda Aleeza
31	Completion	9 d	Malaika Junaid; Syeda Aleeza
32	Project Maintenance & Polishing	9 d	Malaika Junaid; Syeda Aleeza
33	Evaluation & Defense Preparation	9 d	Malaika Junaid; Syeda Aleeza

WBS with Gantt Chart

Table 7 Gantt Chart for Seevia

